

Starting Small with AI: Building Trustworthy KPI Dashboards

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github.com/ElmYgl/kpi_readiness_10step

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Abstract

Institutional research offices routinely deploy KPI dashboards before the underlying metrics have been validated for definition clarity, data quality, governance completeness, or alignment with industry standards. This paper presents a governance-first AI validation framework that addresses this gap directly. Implemented as a three-module open-source Python pipeline (MIT License, Copyright© 2026 Elmer Yglesias), the framework validates any institutional KPI against 10 rigorous steps spanning three phases: Define (metadata registration, definition clarity, entity coherence), Validate (data quality screening, AI use assessment, governance completeness, readiness decision), and Enhance (design quality assessment using a dual SMART and Tufte framework, and industry best practice alignment against EDUCAUSE, AGB, and NACUBO standards).

The core validation engine runs on pure Python with no external dependencies. Two optional modules use the Claude API for plain-language KPI intake and AI-suggested refinements with board-ready output. The framework was piloted at Norco College, where it revealed concrete display failures: a 44-percentage-point pipeline loss hidden by separate display pages, a 32-point livable-wage collapse left unannotated, and equity gaps invisible in tabular form. A live validation run on First-Year Retention Rate demonstrates end-to-end functionality. Freely available at github.com/ElmYgl/kpi_readiness_10step.

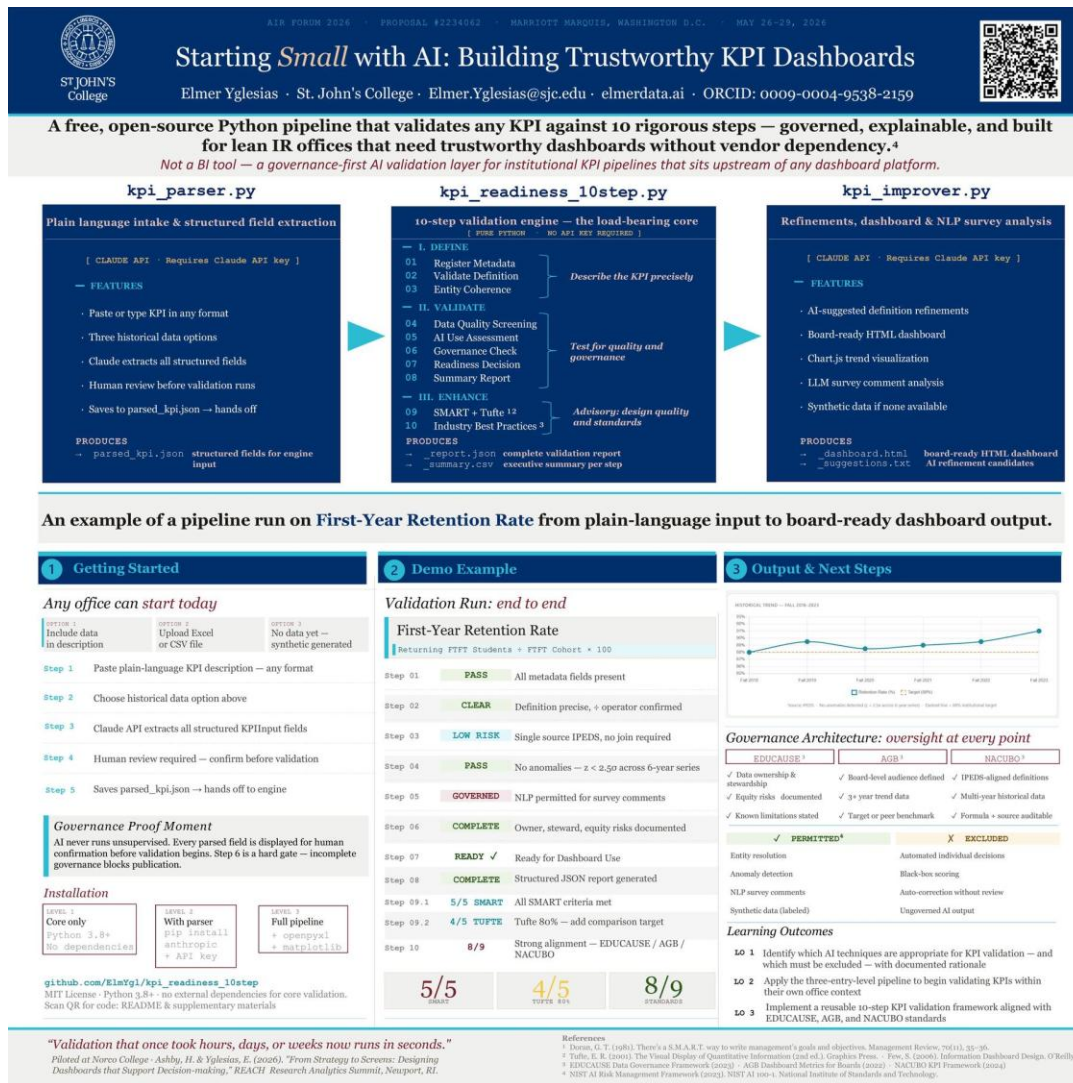


Figure 1. AIR Forum 2026 Poster: Starting Small with AI: Building Trustworthy KPI Dashboards (v62)

1. Introduction

A persistent bottleneck in institutional research is not the absence of dashboard technology. It is the absence of validated, trustworthy KPIs to populate those dashboards. Institutional research offices frequently invest significant effort in visualization platforms while the underlying metrics remain ambiguous in definition, undocumented in governance, and unvalidated for data quality. The result is dashboards that are visually polished but analytically unreliable.

For the purposes of this paper, a trustworthy KPI is defined as a metric whose definition, provenance, governance, interpretive context, and visualization structure are sufficiently documented and validated to support reliable institutional decision-making. The challenge of achieving that standard is well-documented in the IR literature. Swing, Jones, and Ross (2016) found significant variation in IR office capacity and data governance maturity across institutions, with smaller offices disproportionately resource-constrained. McGuire (2017) identified data quality and governance as pressing concerns for IR professionals, noting that the human element remains essential in maintaining data integrity. More recently, Teshome (2025) conducted a systematic review of KPI frameworks in higher education and concluded that persistent implementation challenges, including data reliability issues and the absence of standardized validation frameworks, represent a significant gap in current practice. As AI tools become increasingly embedded in IR workflows, Webber (2023) and the EDUCAUSE 2024 AI Landscape Study have each noted that governance frameworks for AI use in institutional research remain underdeveloped, with fewer than one in four institutions aware of a formal AI policy (EDUCAUSE, 2024).

The visualization literature points to an equally important gap. Tufte (2001) established that graphical integrity is not aesthetic but epistemic: visual representations must be proportional to the data they encode, must show comparisons clearly, and must integrate annotation with data rather than relegating methodology to footnotes. Few (2006) catalogued thirteen common dashboard design mistakes that undermine interpretive accuracy, including inadequate context (metrics displayed without benchmark lines or target overlays), inaccurate encoding (truncated axes and 3D effects that misrepresent magnitude), and poor arrangement (related KPIs on separate pages that obscure meaningful relationships). These principles have rarely been operationalized in IR practice as measurable validation criteria. Step 9 and Step 10 of the framework address that gap directly.

The framework was developed in collaboration with Norco College (Riverside Community College District, Inland Empire, California), a community college serving approximately 15,000 students annually with a 67% Pell-eligible population. The Norco pilot documented specific, concrete failures in existing production dashboards across three high-risk KPI domains, findings that directly motivated the governance and validation architecture presented here. The pilot work was presented at the 2026 REACH Research Analytics Summit (Ashby & Yglesias, 2026); the full presentation is available at the GitHub repository listed above.

The framework is not a business intelligence tool. It is a validation layer that sits upstream of any dashboard platform. Its core intellectual contribution is the reorientation from “What

do we measure?” to “How do we ensure correct interpretation?” (Ashby & Yglesias, 2026). The Python implementation is evidence of the architecture's feasibility, not the architecture itself. Human oversight, bias documentation, equity risk assessment, and hard governance gates are embedded in the workflow at every step.

2. The Problem: KPI Trustworthiness Precedes Visualization

Higher education institutions increasingly invest in dashboard platforms such as Tableau, Power BI, and custom web applications as the primary vehicle for communicating institutional performance to boards, leadership, and accreditors. These platforms are mature, capable, and widely adopted. The bottleneck is not visualization. It is the trustworthiness of the KPIs being visualized and the governance of how they are displayed.

2.1 Evidence from the Norco College Pilot

The Norco College pilot examined three high-risk KPI domains from the college's 2025–2030 Strategic Planning and Governance Manual and found concrete evidence of how display failures hide institutional realities from decision-makers (Ashby & Yglesias, 2026).

CTE wage outcomes (KPIs 5.1–5.2, Goal 5: Workforce and Economic Development) were tracked in the existing Power BI dashboard as bar charts showing median annual earnings and livable-wage attainment rates. The source dashboard displayed these numbers without the MIT Living Wage Calculator benchmark for Riverside County, approximately \$46,000 per year for a single adult, making it impossible for leadership to assess whether program outcomes represented economic mobility or stagnation. More critically, KPI 15 (percent of students attaining a livable wage) showed a collapse from 64% in 2021–22 to 39% in 2022–23, against a projected target of 71%, a 32-point gap that was completely unannotated in the source dashboard. Following Few's (2006) Mistake #2 (inadequate context: bars without benchmark lines), the existing visualization reported a number rather than an evidence-based judgment.

Cohort completion and persistence rates (KPIs 2.3–2.4, Goal 2: Guided Pathways) were displayed on separate dashboard pages, violating Few's (2006) Mistake #9 (poor arrangement: related KPIs not adjacent). Side by side, the two metrics reveal an institutional pipeline story: 61% of first-time students persist to the second term, but only 17% complete a degree or certificate within three years. That 44-percentage-point gap represents the students who persist but do not complete, the core challenge of Guided Pathways implementation. The display structure was hiding this institutional story entirely.

Equity gap metrics (KPIs 3.1–3.2, Goal 3: Eliminate Equity Gaps by 2030) were reported as tabular PPG-1 data rather than visualizations. The tables showed that African-American students had a Math and English completion rate of 9%, which is 8.1 percentage points below the institutional average of 16.3%, requiring an additional 39 students to close the gap. Hispanic students had a completion rate of 15%, or 3.2 percentage points below average, requiring an additional 144 students to reach parity. Visualizing these gaps as dot plots rather than tables revealed both the magnitude and

the number of students needed to close each gap, information that tables cannot communicate at a glance. The pattern illustrates Tufte's (2001) principle of small multiples and Few's (2006) Mistake #10 (ineffective highlighting: color used decoratively rather than to signal threshold breaches).

2.2 Systematic Failure Modes

The Norco findings are not idiosyncratic to one institution. They represent a class of display failures that any IR office can produce when KPIs are published without a validation layer upstream of the visualization platform. Systematic failure modes include:

- Ambiguous definitions that produce inconsistent counts across reporting cycles
- Formulas without documented numerators, denominators, or census dates
- Cross-system joins without specified identifier fields, creating entity resolution risks
- Anomalies in historical data that have never been reviewed or explained
- Governance gaps: missing data owners, undocumented limitations, no equity risk assessment
- Display without context: metrics published without benchmark lines, targets, or comparison periods
- AI use cases applied without documented rationale or exclusion criteria

Each of these failure modes produces a dashboard that is visually credible but analytically unreliable. The governance-first framework presented here addresses each failure mode systematically before any KPI is published to a dashboard audience.

3. Methods: Framework Design and Development

3.1 Design Rationale

The 10-step validation framework was designed around a single organizing principle: governance must precede publication. Every design decision follows from this principle. Steps are ordered so that a KPI cannot advance to quality assessment without first passing metadata completeness, definition clarity, entity coherence, data quality screening, and governance checks. Steps 9 and 10 are intentionally advisory because design quality and industry alignment are improvement targets, not binary pass/fail gates.

The choice of 10 steps reflects the IR office workflow, not an arbitrary round number. Steps 1–3 address what must be true before any data quality work can be done: the KPI must be completely defined, its formula must be unambiguous, and its data provenance must be traceable. Steps 4–6 address the three questions an IR director must be able to answer before any KPI goes to a dashboard: Is the historical data clean? Are AI uses documented and bounded? Is governance complete? Step 7 is the readiness gate that synthesizes those answers. Step 8 produces the audit trail. Steps 9–10 produce the advisory scorecard.

The SMART framework (Doran, 1981) was chosen for Step 9.1 because it is the most widely recognized KPI design standard in management practice and provides a common vocabulary for IR practitioners and institutional leaders. Tufte's visualization principles (Tufte, 2001) were chosen for Step 9.2 because they address the specific failure modes documented at Norco: absence of comparison context, inaccurate encoding, and insufficient data density. Few's (2006) dashboard design safeguards informed the selection of Tufte criteria but were not implemented as a separate scoring framework; they are embedded in the rationale for each Tufte criterion as concrete institutional examples. The EDUCAUSE, AGB, and NACUBO standards in Step 10 were selected because they represent the three governance bodies whose frameworks IR directors are most likely to cite to institutional leadership.

3.2 Validation Criteria Design

Each step's validation criteria were designed to be transparent, reproducible, and adjustable by any office. Transparency means an IR director should be able to read the Python source code for any step and understand exactly what it checks and why. Reproducibility means two analysts running the same KPI through the framework on the same data should get the same result. Adjustability means an office with different thresholds can modify the named constants at the top of the file without touching the logic.

The outlier detection threshold of $z \geq 2.5\sigma$ (Step 4) was chosen as a conservative flag rather than a definitive anomaly designation. The framework never characterizes a flagged data point as an error; it characterizes it as requiring human review. The 25% step-change threshold was chosen to catch period-over-period changes large enough to represent either a data quality issue or a policy change requiring annotation, exactly the kind of unannotated collapse documented in the Norco CTE wage data.

The governance hard gate (Step 6) was designed with a single non-negotiable requirement: `human_signoff_required` must be `True`. This field cannot be left blank, cannot default to `False`, and cannot be overridden by any other step. Its purpose is not bureaucratic. It is architectural. Making human signoff a required field forces every KPI through a human decision point before the validation report can pass.

3.3 Norco College Pilot

The framework was piloted at Norco College between January and April 2026 in collaboration with Hayley Ashby, Ed.D. (Dean, Institutional Effectiveness). The pilot applied the framework's design principles to 13 KPIs from the college's 2025–2030 Strategic Planning and Governance Manual, mapped to 8 strategic goals and organized across a 9-stage dashboard development roadmap (Purpose, Metrics, Data, Model, Design, Build, Validate, Deploy, Improve). The Norco planning context is embedded in a multi-level governance hierarchy spanning the California Community Colleges Vision 2030, the RCCD 2025–30 Strategic Plan, and Norco's 10-year Educational Master Plan.

The pilot focused on Stages 5 (Design) and 7 (Validate) of the roadmap, the two stages carrying the highest interpretation risk, and selected three KPI domains for in-depth analysis based on interpretive complexity and potential for display-driven misrepresentation: CTE wage mapping (highest benchmark integration complexity),

cohort completion and transfer (highest temporal complexity due to 3-year lag), and equity gap metrics (highest visual-distortion risk). For each domain, the pilot documented failure modes in the existing Power BI dashboard, applied Tufte and Few design principles to redesign the visualization, and identified the governance documentation required to make the redesigned display auditable.

The pilot produced preliminary outcome data that informed framework development. Of the 13 KPIs reviewed, 10 lacked explicit comparison benchmarks, satisfying neither the Tufte data density criterion nor the AGB board governance standard. Seven lacked fully documented governance ownership across at least one required field. Five contained visualization structures, including tabular equity data and separate-page cohort metrics, judged to impair interpretability under Tufte and Few criteria. All three KPI domains required definition refinement to meet the Step 02 clarity threshold. These findings established a baseline: in a well-resourced California community college with an active institutional effectiveness function, the majority of strategic KPIs were not dashboard-ready under the framework's criteria before remediation. The pilot findings directly informed three design decisions in the framework. First, the Tufte data density criterion (Step 9.2) was refined to explicitly check for comparison context because every KPI examined at Norco failed interpretability without it. Second, the AGB benchmark standard (Step 10) was added because the Norco audit demonstrated that even well-governed KPIs can lack the comparison context that board-level audiences require. Third, the convergent finding, in which the same gap (missing benchmark) is independently flagged by Tufte (Step 9.2) and AGB (Step 10), was recognized as a validation feature rather than a redundancy: two independent theoretical frameworks identifying the same improvement opportunity strengthens the recommendation.

4. The Governance-First Validation Framework

4.1 Architecture Overview

The framework consists of three independent Python modules that can be used separately or in sequence (see Appendix D for installation and quick-start instructions):

kpi_parser.py	Plain-language intake module. Accepts a KPI description in any format, uses the Claude API to extract structured fields, displays all parsed fields for human review, and hands off to the validation engine. Requires Claude API key.
kpi_readiness_10step.py	Core validation engine. Runs 10 validation steps on pure Python with no external dependencies. No API key required. The load-bearing center of the framework.
kpi_improver.py	Refinement and dashboard module. Reads the validation report, uses the Claude API to generate suggested improvements, produces a board-ready HTML dashboard with Chart.js trend

	visualization, and performs LLM-powered survey comment analysis.
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The term ‘pipeline’ describes the full sequential flow when all three modules are used together, but each module operates independently. The Claude API calls in `kpi_parser.py` and `kpi_improver.py` could be replaced with any LLM API by updating the client initialization. The framework is LLM-agnostic at its core.

Note on the demonstration run: the appendix outputs were generated by running `kpi_readiness_10step.py` and `kpi_improver.py` directly on the built-in demo KPI. `kpi_parser.py` was not invoked for this run; it is used when starting from a plain-language description. In a full pipeline run, `kpi_parser.py` would accept a plain-language description, extract all structured fields via the Claude API, display them for human review, and hand off automatically.

4.2 Three Entry Levels

The framework is designed for accessibility across offices with varying technical capacity and AI budgets (see Appendix D for setup instructions at each level):

Level 1: Core only	<code>kpi_readiness_10step.py</code> · Python 3.8+ · No external dependencies · No API key required · Any office can start today
Level 2: With parser	<code>kpi_parser.py</code> + validation engine · <code>pip install anthropic</code> · Claude API key required · Removes technical friction at intake
Level 3: Full pipeline	All three modules · <code>pip install anthropic openpyxl matplotlib</code> · Complete AI-assisted experience from plain-language input to board-ready dashboard

4.3 The 10-Step Validation Engine

The core validation engine runs 10 sequential steps organized into three phases:

Phase I: Define (Steps 1–3). Establish that the KPI is precisely and completely described before any quality checks are run.

Step 01: Register Metadata	Register	Checks that all required KPIInput fields are present and non-empty. Status: Pass / Needs Attention
Step 02: Definition	Validate	Checks definition length and precision, formula calculation operator, and time period specification. Status: Clear / Ambiguous

Step 03: Coherence	Entity	Determines whether the KPI requires cross-system record joins. Assigns entity risk level (Low / Moderate / High). Status: Pass / Fail
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Phase II: Validate (Steps 4–8). Test the KPI for data quality, appropriate AI use, governance completeness, and publish readiness.

Step 04: Data Quality Screening	Analyzes historical aggregates for outliers ($z\text{-score} \geq 2.5\sigma$) and step changes ($\geq 25\%$ period-over-period). Every anomaly is flagged for human review. The tool never auto-corrects. Status: Pass / Needs Human Review
Step 05: AI Use Assessment	Documents permitted AI techniques (entity resolution, anomaly detection, NLP for survey comments, labeled synthetic data) and categorical exclusions (automated individual decisions, black-box scoring, auto-correction without review). Status: Pass (documents, not blocks)
Step 06: Governance Check	Verifies data owner, data steward, known limitations, bias/equity risks, and human signoff required (must be True). Incomplete governance is a hard blocker. Status: Complete / Incomplete
Step 07: Readiness Decision	Makes the final publish readiness decision based on Steps 2, 3, 4, and 6. Four blockers prevent dashboard publication: ambiguous definition, high entity risk, unreviewed anomalies, incomplete governance. Status: Ready for Dashboard Use / Not Ready
Step 08: Summary Report	Compiles a structured one-page summary exported to JSON and CSV (see Appendix A). Optionally generates a time-series PNG chart via matplotlib. Status: Generated

Step 05 also governs the framework's approach to natural language processing. `kpi_readiness_10step.py` includes a keyword-based survey comment categorizer (`demo_nlp_survey_comments()`) using transparent string matching with no AI dependency; classification rules are readable directly in the source code. `kpi_improver.py` contrasts this with a full LLM-powered analysis using the Claude API, returning sentiment labels, priority flags, and a key insight. The two approaches are displayed side-by-side in the HTML dashboard output and explicitly labeled by method, demonstrating that AI augmentation should be transparent and comparable to its non-AI baseline.

Phase III: Enhance (Steps 9–10). Advisory assessments of design quality and industry alignment. These steps do not affect the Step 07 readiness decision.

Step 9 is the paper's most distinctive methodological contribution: a dual-framework assessment that evaluates KPI quality from two independent angles simultaneously. Step 09.1 applies the SMART framework (Doran, 1981) to assess internal design quality, specifically whether the KPI is specific, measurable, achievable, relevant, and time-bound. Step 09.2 applies Tufte's five visualization principles (Tufte, 2001) to assess external dashboard readiness. Running both assessments on every KPI makes a claim that neither alone can make: a KPI that passes both is well-defined and visualization-ready. SMART governs what the metric is; Tufte governs how it will be seen.

Framework	Governs
SMART (Doran, 1981)	Metric construction: internal design quality. Is the KPI specific, measurable, achievable, relevant, and time-bound?
Tufte/Few (Tufte, 2001; Few, 2006)	Metric interpretation: external visualization readiness. Will the KPI chart clearly, honestly, and with sufficient context?
Governance layer (Steps 1–6)	Institutional accountability: is ownership documented, governance complete, and human signoff required before publication?

These three layers are intentionally independent. A KPI can pass SMART and fail Tufte (well-defined but visualization-poor). It can pass both and fail governance (well-designed but ungoverned). The framework requires all three layers before a KPI is considered dashboard-ready.

Step 09.1: SMART Assessment	Evaluates the KPI against five SMART criteria: Specific (definition precision), Measurable (formula operator), Achievable (historical variance $\leq 30\%$), Relevant (audience defined), Time-bound (reporting cycle specified). Score: X/5
Step 09.2: Tufte Assessment	Evaluates visualization readiness against five Tufte principles: Data-ink ratio (KPI is a ratio with time series), Chartjunk avoidance (simple units and clear direction), Data density (comparison context: target or benchmark), Small multiples readiness (can be disaggregated by demographic group), Narrative clarity (clear ownership and actionable direction). Score: X/5 (XX%)
Step 10: Industry Best Practices	Checks alignment to EDUCAUSE (3 standards), AGB (3 standards), and NACUBO (3 standards). Score: X/9

5. Governance Architecture

5.1 The Governance Proof Moment

The most important governance mechanism in the framework is the human review checkpoint in `kpi_parser.py`. After the Claude API parses a plain-language KPI description into structured fields, every parsed field is displayed to the user before validation runs. The user must explicitly confirm (Y) to proceed. AI never runs unsupervised. The

deliberate pause is the governance proof moment: the point at which human judgment is formally inserted into the AI-assisted workflow.

Step 06 enforces a second hard gate. If data owner, data steward, known limitations, bias/equity risks, or human signoff required are missing or incomplete, the KPI cannot pass Step 07. Governance completeness is not optional; it is a publish prerequisite.

The Norco College pilot reinforced the importance of this architecture. Ashby and Yglesias (2026) articulated the underlying principle directly:

“Design discipline without governance is temporary. Governance without design standards is incomplete. Both are required.” — Ashby & Yglesias, 2026

This principle is operationalized throughout the framework. The Tufte assessment (Step 09.2) governs display standards, while the governance check (Step 06) and industry best practices assessment (Step 10) enforce the institutional conditions that make principled display sustainable.

5.2 Permitted and Excluded AI Techniques

Permitted Techniques	Excluded Techniques
Entity resolution: to unify records across systems	Automated decisions affecting individuals
Anomaly detection: aggregate screening without auto-correction	Black-box scoring without documentation
NLP for survey comments: keyword-based or LLM-powered	Auto-correction of records without human review
Synthetic data: clearly labeled, for visualization only	Ungoverned AI output published without review

5.3 Industry Standards Alignment

Step 10 assesses each KPI against nine standards from three higher education governance organizations. EDUCAUSE Data Governance Framework (2023): data ownership and stewardship, equity risks documented, known limitations stated. AGB Dashboard Metrics for Boards (2022): board-level audience defined, 3+ year trend data, target or peer benchmark. NACUBO KPI Framework (2024): IPEDS-aligned definitions, multi-year historical data, formula and source auditable. The framework is also grounded in the NIST AI Risk Management Framework (2023), which informs the AI use assessment and governance architecture throughout. A KPI scoring 85% or above receives a Strong Alignment designation.

6. Demonstration: First-Year Retention Rate

The following validation run demonstrates the framework applied to First-Year Retention Rate, defined as the percentage of first-time, full-time degree-seeking undergraduates who return for the following fall term. Formula: Returning FTFT Students divided by FTFT Cohort, multiplied by 100. Primary source: IPEDS. Six years of historical data (Fall 2018 through Fall 2023). The run was executed directly using `kpi_readiness_10step.py` and `kpi_improver.py` on the built-in demo KPI.

First-Year Retention Rate is representative of the cohort-based KPI types examined in the Norco College pilot, specifically metrics where temporal complexity, benchmark context, and equity disaggregation all carry interpretive risk. The validation run below demonstrates how the framework operationalizes the governance and display standards developed in that pilot context.

Step 01	PASS: All metadata fields present
Step 02	CLEAR: Definition precise, division operator confirmed
Step 03	LOW RISK: Single source IPEDS, no join required
Step 04	PASS: No anomalies detected ($z < 2.5\sigma$ across 6-year series)
Step 05	GOVERNED: NLP permitted for survey comments
Step 06	COMPLETE: Owner, steward, equity risks documented
Step 07	READY ✓: Ready for Dashboard Use
Step 08	COMPLETE: Structured JSON report and time-series chart generated
Step 09.1	5/5 SMART: All SMART criteria met
Step 09.2	4/5 TUFTE: 80% visualization readiness; add comparison target
Step 10	8/9: Strong Alignment (89%) across EDUCAUSE, AGB, and NACUBO

The full pipeline produces five output files. `kpi_readiness_10step.py` generates three: `_report.json` (complete 10-step audit trail), `_summary.csv` (executive summary, reproduced in Appendix A), and `_chart.png` (matplotlib time-series with anomalies marked). `kpi_improver.py` generates two additional files: `_dashboard.html` (board-ready HTML dashboard with Chart.js trend visualization and LLM survey analysis, see Appendix C) and `_suggestions.txt` (AI-drafted refinements for human review, excerpt in Appendix B). Installation and run instructions are in Appendix D.

Two findings from the Enhance phase are worth noting. Step 09.2 scored 4/5 on Tufte principles because the KPI lacks a comparison target or peer benchmark, failing the data density criterion. Step 10 scored 8/9 because the AGB benchmark requirement independently flags the same gap from a governance perspective. These two assessments, one from visualization theory and one from industry standards, converge on the same actionable recommendation: add a peer comparison target before publishing

to a board audience. This convergence is not coincidental. It is a validation feature: the same improvement is required by both Tufte's evidence-centered display principles and AGB's board governance standards. The finding mirrors the central lesson of the Norco pilot: every KPI examined, including CTE wages, cohort completion, and equity gaps, failed interpretability without comparison context (Ashby & Yglesias, 2026).

7. Implications for Institutional Research Practice

Accessibility for lean offices. The core validation engine requires only Python 3.8+ and no external dependencies. An office with no AI budget and no dedicated technical staff can download `kpi_readiness_10step.py` and validate KPIs today. The three-entry-level design ensures that AI augmentation is additive, not prerequisite. See Appendix D for step-by-step setup instructions.

Separation of validation from visualization. The framework establishes a clear architectural distinction: validation happens upstream of any dashboard platform. The separation protects the integrity of published dashboards regardless of which visualization tool an institution uses. As the Norco pilot demonstrated, the problem is not the platform. Power BI is capable of displaying the MIT Living Wage benchmark and the persistence-to-completion pipeline. The problem is the absence of validated, governed display standards upstream of the platform (Ashby & Yglesias, 2026).

Governance as infrastructure. By embedding data owner, data steward, equity risk, and human signoff requirements as hard prerequisites for publication, the framework operationalizes governance rather than treating it as abstract compliance language. The governance proof moment in the intake module makes human oversight tangible and auditable.

Equity monitoring by design. The bias and equity risk field is a required governance element. The Tufte small multiples assessment checks whether the KPI can be disaggregated by demographic group. The Norco pilot demonstrated why this matters: equity gap metrics carry the highest visual-distortion risk in any KPI portfolio (Ashby & Yglesias, 2026). At Norco, the difference between a table and a dot plot was the difference between a number and an actionable equity gap: 9% completion for African-American students, 8.1 percentage points below average, with 39 students needed to close the gap.

Constrained AI as a trust signal. The permitted/excluded AI table, which prominently features automated individual decisions, black-box scoring, and auto-correction without review as categorical exclusions, communicates methodological restraint to an IR audience increasingly skeptical of AI overclaiming. Responsible boundaries are a feature, not a limitation.

8. Limitations and Future Directions

Single-institution pilot. The framework has been piloted at Norco College, one institution with a specific governance context: a California community college aligned with RCCD, Vision 2030, and the Guided Pathways framework, serving a 67% Pell-eligible population of approximately 15,000 students annually. Multi-institution validation across diverse SIS environments, IPEDS reporting structures, four-year institutions, and different governance cultures has not yet been conducted. That is the critical next empirical step.

Synthetic demonstration data. The First-Year Retention Rate demonstration run uses synthetic data built into the framework. The six-year series (Fall 2018 through Fall 2023) was designed to represent plausible retention patterns for a small college, but it is not institutional data. The Norco pilot involved real KPIs from real production dashboards; the demonstration in this paper does not. Readers who wish to run the framework on real data should use their institution's own historical aggregates, as described in Appendix D.

Heuristic validation rules. The validation steps use transparent heuristics: minimum definition length, presence of calculation operators, and z-score thresholds for outlier detection. Explainability was chosen over statistical sophistication deliberately, so that an IR director can read the code and understand every decision the tool makes. Offices working with specialized KPI types such as financial ratios, weighted composite indices, or non-standard reporting cycles will need to adjust thresholds, which are documented as named constants in the source code.

API dependency for full functionality. Two of the three modules require a Claude API key. The core validation engine runs without one. The framework was designed to be LLM-agnostic: the Claude API calls can be replaced with OpenAI, Gemini, Azure OpenAI, or local models by updating the client initialization.

Future directions include multi-institution validation studies as the immediate priority, integration with common SIS platforms (Banner, Ellucian, Workday Student), a natural language interface that removes the command-line requirement, and a web-based front end for non-technical users. The framework is developed in the open under an MIT License; collaboration from IR offices willing to test it against their own KPIs is actively welcomed.

9. Conclusion

The governance-first AI validation framework presented here addresses a real and underserved problem in institutional research: the gap between KPI deployment and KPI trustworthiness. The Norco College pilot demonstrated that this gap is not hypothetical. It exists in live production dashboards at a real institution, producing concrete failures: a 32-point livable-wage attainment collapse left unannotated in the source dashboard, a 44-percentage-point pipeline loss hidden by separate display pages, and equity gaps visible only as dot plots that tables could not reveal.

Taken together, these design choices offer a practical path toward more reliable institutional dashboards: governance as prerequisite, bounded AI, dual-framework quality assessment, and zero-cost core access.

The framework's intellectual contribution is not the Python code itself. It is the governance architecture: the operationalization of human oversight, the formalization of KPI readiness as a measurable state, and the demonstration that AI can be used responsibly in institutional research without sacrificing explainability, equity awareness, or accountability.

“Better dashboards don’t show more data. They show the right evidence, the right way.” — Ashby & Yglesias, 2026

Validation steps that previously had no systematic framework now run in seconds. The remaining work is not technical. It is institutional: getting IR offices to validate before they publish, to document before they deploy, and to govern before they visualize. That is the problem this framework is designed to help solve.

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Appendices

The following appendices present actual output files generated by running `kpi_readiness_10step.py` and `kpi_improver.py` on the First-Year Retention Rate demo KPI. Appendix D provides installation and quick-start instructions for all three entry levels.

Appendix A. Executive Summary Output (`_summary.csv`)

Generated by `kpi_readiness_10step.py`. Opens directly in Excel. Primary deliverable for IR directors and data stewards.

KPI VALIDATED	
KPI Name	First-Year Retention Rate
Formula	Returning FTFT Students / FTFT Cohort
Primary Source	IPEDS / SIS
READINESS RESULTS	
Readiness Status	Ready for Dashboard Use
Anomalies Detected	No
Design Quality	Strong Design and Visualization-Ready (5/5 SMART, 4/5 Tufte 80%)
Industry Standards	Strong Alignment (8/9, 89%)
GOVERNANCE	
Data Owner	Registrar
Data Steward	Institutional Research
Human Signoff	Required (True)
Known Limitations	COVID-era cohorts may show atypical patterns; census date timing affects counts
Bias/Equity Risks	Retention rates vary by demographic segment; monitor equity gaps by race/ethnicity and income level

Appendix B. AI-Suggested Refinements Excerpt (`_suggestions.txt`)

Generated by `kpi_improver.py`. All content is AI-generated and requires human review and data steward approval before adoption.

AI-GENERATED: Human review and data steward approval required before adoption

Suggested Definition Refinement. Consider adding "at the same institution" and "measured at census date" to clarify that transfers out are counted as non-retained and to specify the measurement point. Suggested: "Percentage of first-time, full-time degree-seeking undergraduates from a fall cohort who re-enroll at the same institution for the subsequent fall term, measured at census date."

Suggested Governance Addition: Target/Goal. The validation flagged missing comparison context (Tuft data density criterion, Step 09.2; AGB benchmark standard, Step 10). Recommended options include a peer benchmark (meet or exceed peer median), an aspirational peer goal (reach peer average by a target year), or a maintenance threshold (maintain above institutional floor). A peer comparison target provides external validation without requiring justification of an arbitrary number.

Suggested Next Steps. (1) Confirm peer group and document selection rationale. (2) Establish target through appropriate governance. (3) Prepare disaggregated equity data by race/ethnicity, Pell status, and first-generation status for at least 3 cohort years. (4) Document census date timing in metadata. (5) Create COVID cohort annotation language for consistent use across reporting.

Appendix C. Historical Trend Chart (_dashboard.html)

The following chart is embedded in the board-ready HTML dashboard generated by `kpi_improver.py`. It shows the six-year retention series (Fall 2018 through Fall 2023) with a dashed institutional target line at 88%. Produced via Chart.js and renders in any browser.

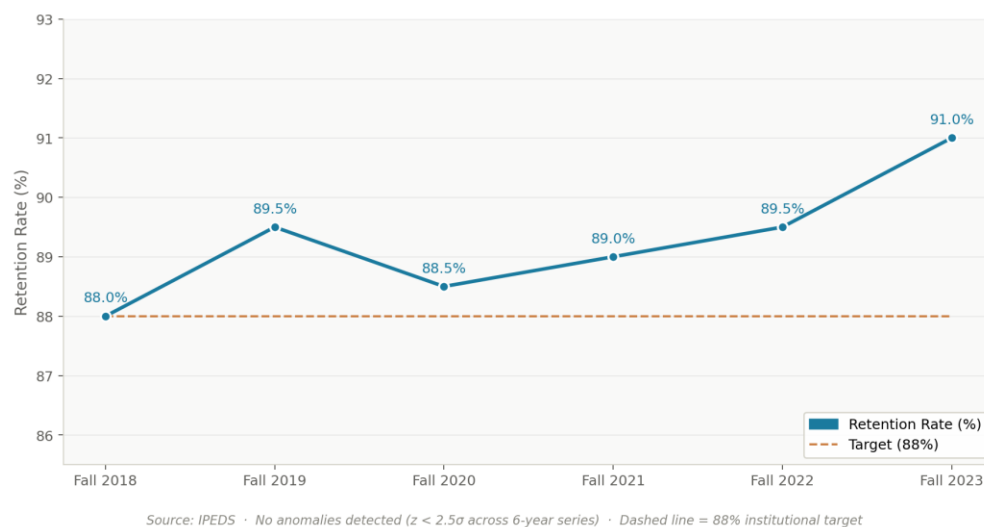


Figure 2. Historical trend chart, First-Year Retention Rate, Fall 2018 through Fall 2023. Source: IPEDS synthetic demo data. No anomalies detected ($z < 2.5\sigma$). Dashed line = 88% institutional target. Generated by `kpi_improver.py` via Chart.js.

Appendix D. Installation and Quick-Start Guide

The following instructions cover all three entry levels. Full README available at github.com/ElmYgl/kpi_readiness_10step.

Prerequisites

All levels require Python 3.8 or later. To check your version:

```
| python --version
```

No other prerequisites are required for Level 1.

Level 1: Core Validation Only (No API Key Required)

Download `kpi_readiness_10step.py` from GitHub and run it directly:

```
| python kpi_readiness_10step.py
```

Generates `_report.json` and `_summary.csv`. To validate your own KPI, edit the KPIInput block in the `main()` function.

Level 2: With Parser (Claude API Key Required)

Install dependencies and set API key:

```
| pip install anthropic
```

Windows:

```
| [System.Environment]::SetEnvironmentVariable('ANTHROPIC_API_KEY', 'your-key-here', 'User')
```

Mac / Linux:

```
| export ANTHROPIC_API_KEY='your-key-here'
| python kpi_parser.py
```

The parser prompts for a KPI description, historical data option, and human review confirmation, then runs `kpi_readiness_10step.py` automatically.

Level 3: Full Pipeline (All Three Modules)

Install all dependencies:

```
| pip install anthropic openpyxl matplotlib
| python kpi_parser.py
| python kpi_improver.py
```

`kpi_improver.py` automatically finds the most recent `_report.json` and generates `_dashboard.html` and `_suggestions.txt`.

Output Files Reference

<code>_report.json</code>	Complete 10-step audit trail with full KPI identity block. For IR staff, archival, and GitHub.
<code>_summary.csv</code>	Executive one-page readiness snapshot. Opens in Excel. For IR directors and data stewards.
<code>_chart.png</code>	Time-series visualization with anomalies marked. Generated by matplotlib. For IR staff and reports.
<code>_dashboard.html</code>	Board-ready HTML dashboard with Chart.js trend visualization and LLM survey analysis. For leadership.
<code>_suggestions.txt</code>	AI-drafted definition, governance, and next-steps refinements. For data stewards.

About the Author

Elmer Yglesias is Chief Data Officer at St. John's College. His work focuses on governance-first data infrastructure, KPI trustworthiness, and responsible AI adoption in institutional research. Contact: Elmer.Yglesias@sjc.edu or elmerdata.ai.

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Freely available at github.com/ElmYgl/kpi_readiness_10step. The repository includes:

- `kpi_parser.py`: plain-language intake module (Claude API)
- `kpi_readiness_10step.py`: 10-step validation engine (pure Python, no external dependencies)
- `kpi_improver.py`: refinement and dashboard module (Claude API)
- README with installation instructions at all three entry levels
- Sample outputs from the First-Year Retention Rate validation run: `_report.json`, `_summary.csv`, `_chart.png`, `_dashboard.html`, `_suggestions.txt`
- `REACH_2026_Strategy_to_Screens_Ashby_Yglesias.pdf`: full 36-slide presentation from the REACH Research Analytics Summit (April 2026) documenting the Norco College pilot. Posted with permission of co-author Hayley Ashby, Ed.D., Norco College.